2D GAN with Local and Global Learning for the Dehazing Echocardiography Challenge in DehazingEcho2025

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Abstract. Echocardiography is crucial for diagnosing cardiovascular diseases but often suffers from image quality issues due to noise, which can impede accurate clinical diagnosis. To address this challenge, we propose a novel algorithm designed to enhance the quality of echocardiographic images, particularly for patients who are difficult to image, thereby improving diagnostic accuracy and the effectiveness of AI-assisted diagnostic tools. In this work, we introduce a 2D Generative Adversarial Network (GAN) framework that combines a local and global learning strategy for dehazing echocardiography. This framework is trained through a process of one-sided image translation. Specifically, the 2D GAN model employs a Mamba-enhanced Symmetric UNet (MSUNet) as the generator, which integrates local and global learning capabilities, and utilizes a PatchGAN as the discriminator. The spatially-correlative loss is used to optimize the 2D GAN model, which can effectively capture spatial relationships within an image rather than domain appearance. An Adam optimizer with a learning rate of 1e-4 is employed, along with a linear learning rate scheduler. Additionally, the input images are resized to 256×256 pixels, and the model is trained with a batch size of 8. The results on the final test dataset validated the effectiveness of the proposed model.

Keywords: Dehazing echocardiography, 2D GAN, Local and global learning, Mamba-enhanced Symmetric UNet.

1 Introduction

Echocardiography has revolutionized cardiac imaging by offering real-time capabilities and cost-effectiveness, enabling cardiologists to assess the basic functioning of the heart and detect critical abnormalities indicative of cardiovascular diseases. However, transthoracic echocardiograms often suffer from various noise sources that degrade image quality and limit interpretability. This issue is particularly pronounced in patients

who are more difficult to image, where the noise can significantly hinder accurate diagnosis. To address this challenge, the Dehazing Echocardiography Challenge 2025 aims to enhance echocardiographic image quality through advanced computational algorithms, focusing on improving the accuracy of medical diagnoses and the performance of AI-assisted algorithms for these challenging cases.

In this challenge, the dataset from easy-to-image and difficult-to-image patients are non-paired, and computational resources are limited. To address these challenges, we have implemented a one-sided image translation approach using a 2D Generative Adversarial Network (GAN). This innovative algorithm is tailored to enhance image quality efficiently, ensuring that our algorithm can effectively dehaze echocardiographic images while operating within the given constraints. Additionally, considering the stability and generalization capabilities of GAN models, we have chosen the architecture proposed in [1] as the foundation for our solution.

However, echocardiography images from difficult-to-image patients often exhibit significant noise and low contrast, which further complicates the dehazing process. To address these issues, we design a novel generator based on the Mamba-enhanced Symmetric UNet (MSUNet). This architecture integrates both local and global learning capabilities, allowing for more effective feature extraction and image enhancement. The MSUNet leverages the strengths of symmetric encoder and decoder, combined with the skip connection, to improve the overall quality of the dehazed images. This approach ensures that the generator can effectively capture both fine-grained details and broader contextual information, leading to clearer echocardiographic images.

In summary, our proposed model combines the efficiency of one-sided image translation with the local and global learning capabilities of the MSUNet architecture. By addressing the specific challenges posed by difficult-to-image patients, we aim to significantly enhance the diagnostic value of echocardiographic images. Our approach not only improves image clarity but also operates within the constraints of limited computational resources, making it a practical and effective solution for real-world applications.

2 2D GAN for the Dehazing Echocardiography

2.1 Data Processing

The DehazingEcho2025 dataset provides 4376 easy-to-image ultrasound images and 2324 difficult-to-image ultrasound images. In order to increase the number of difficult-to-image ultrasound images and improve the robustness of the model, we performed data augmentation on the difficult-to-image ultrasound images. The augmentation strategies included random resized crop with a scale of 0.8 to 1.0 and a ratio of 0.8 to 1.2, random horizontal flip, random vertical flip, and random rotation within ±20 degrees.

2.2 Model Development

In this section, we present a detailed description of our methodology for dehazing echocardiography. Our model development includes a 2D GAN framework with unique architectures for both the generator and discriminator, trained using spatially-correlative loss function.

As shown in Fig. 1, existing unpaired image-to-image translation methods either employ cycle-consistency loss within a two-sided framework [2, 3] or utilize alternative forms in a one-sided framework [4, 5] to preserve content. To reduce memory consumption and minimize domain-specific peculiarities, we adopt the architecture proposed in [1] as the foundation for our solution, trained through a process of one-sided image translation.

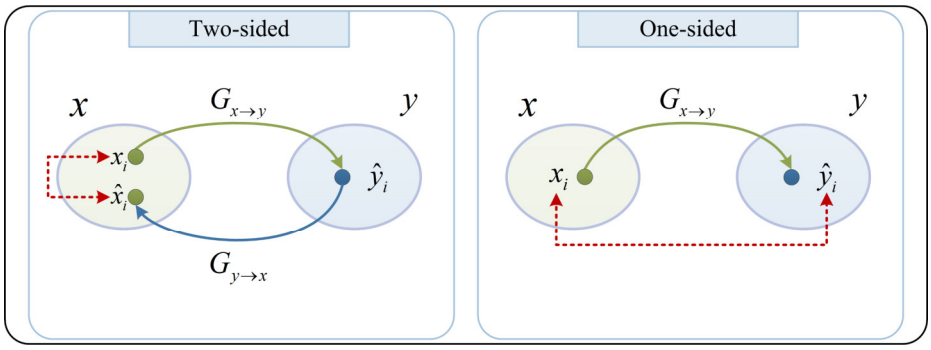

Fig. 1. Comparison of unpaired image-to-image translation methods.

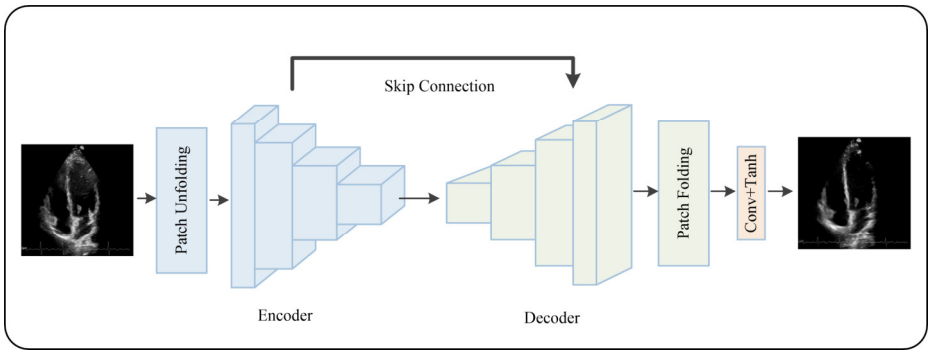

Fig. 2. Overview of the proposed generator architecture in 2D GAN: Mamba-enhanced Symmetric UNet (MSUNet). The network consists of Encoder, Decoder and Skip Connection.

Fig. 2 presents our generator network, which we call the Mamba-enhanced Symmetric UNet (MSUNet). This network incorporates a local and global learning strategy and consists of three main components: Encoder, Decoder, and Skip Connections. For the GAN's discriminator, PatchGAN [6] is used, which evaluates patches of an image rather than the entire image at once. This approach allows the discriminator to focus on smaller regions, which can help in capturing finer details and improving the quality of generated images.

Given that the generator adopts a symmetric architecture, we provide a detailed illustration of the specific details of the encoder, as shown in Fig. 3. For a noisy ultrasound image, the Patch Unfolding layer divides it into non-overlapping image patches, which are then processed through a 3×3 convolution to obtain the image embeddings. As the features propagate through the encoder, the resolution of the embeddings gradually decreases while the number of channels progressively increases, forming a hierarchical structure. To effectively capture both local and global information, we design a local-global strategy for patch embeddings. Specifically, the Residual Block (ResB) effectively extracts features from the local patch embeddings at each layer. Subsequently, all the local patch embeddings are stacked to form the input sequence for Mamba2D, which models global dependencies in the input. Through this design, our model is able to capture both local details and global structural information simultaneously, thereby better preserving textures and overall consistency in the generated images. This approach not only improves the quality of the generated images but also enhances the model's adaptability to complex scenes.

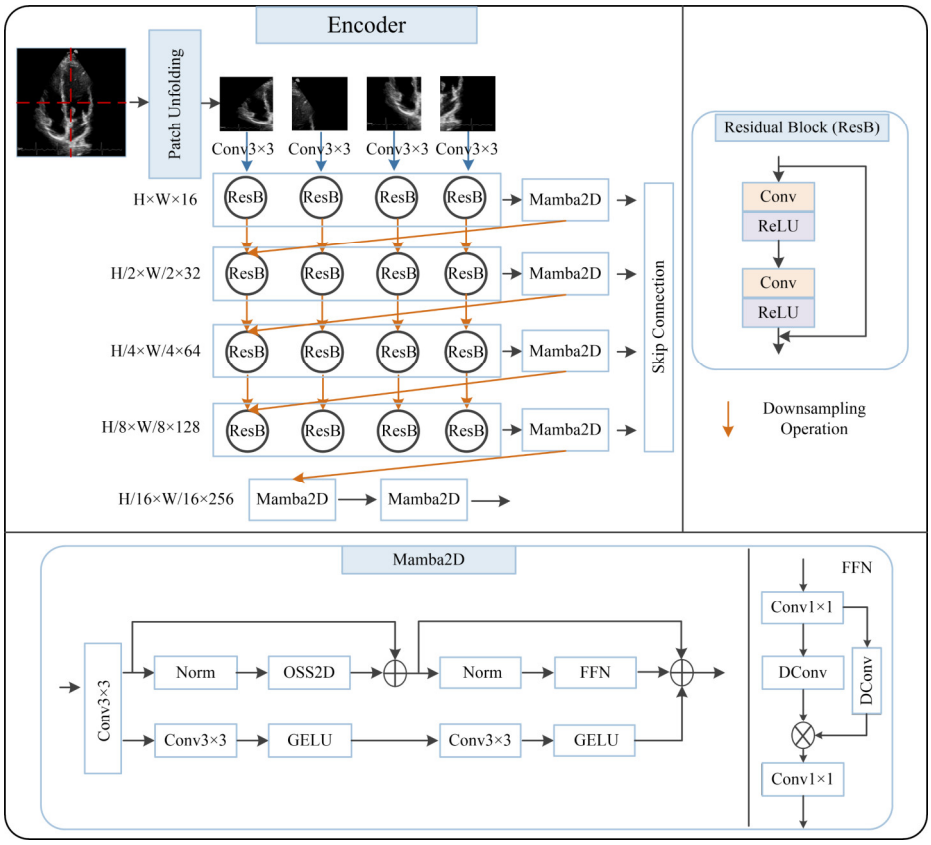

Fig. 3. Overview of the detailed encoder in the proposed generator.

2.3 Loss Function

During the model training process, three primary loss functions are employed: discriminator loss (L_D), spatially-correlative loss (L_S) and generator loss (L_G).

$$L_D = -\mathbb{E}_{y\sim p_d}[logD(y)] - \mathbb{E}_{\hat{y}\sim p_g}\left[log\left(1 - D(\hat{y})\right)\right]$$
$$L_G = \mathbb{E}_{\hat{y}\sim p_g}\left[log\left(1 - D(\hat{y})\right)\right] + \lambda d\left(S_x, S_{\hat{y}}\right) \quad (1)$$

where L_D denotes the adversarial loss for the discriminator (D), $\hat{y}$ is the dehazing image. For detailed information on the spatially-correlative loss (L_S), please refer to [1]. L_G is the loss for the generation network (G), which consists of the style loss term and the structure loss term. The hyperparameter λ is set to 10. S_x and $S_{\hat{y}}$ are spatially-correlative maps in the source domain and the corresponding maps in target domain, respectively. $d(\cdot)$ denotes the cosine distance.

3 Experiments and Analysis

3.1 Experiment settings

After the aforementioned data augmentation, we obtained 4376 easy-to-image ultrasound images and 4648 difficult-to-image ultrasound images. Subsequently, the dataset was divided into training and testing datasets at a ratio of 8:2. For the 2D GAN, the Adam optimizer was used to update the model weights. The initial learning rate was set to 1e-4 with $\beta_1 = 0.5$ and $\beta_2 = 0.999$ for training, complemented by a linear learning rate scheduler. The model was trained for 400 epochs with a batch size of 8. The input image was resized to a resolution of 256×256 pixels, with the patch size set to 64×64. All models were implemented by PyTorch framework equipped with an NVIDIA RTX 5880 Ada Generation GPU with 48GB of memory.

3.2 Experimental results

Fig. 4 presents the dehazing results on the DehazingEcho2025 dataset, showcasing the enhanced clarity and detail in the ultrasound images post-dehazing treatment. The top row illustrates the original difficult-to-image inputs, while the bottom row displays the corresponding dehazing outputs, highlighting the effectiveness of our algorithm in improving image quality. Fig. 5 illustrates the FID metric scores on the test dataset across various epochs, showing a general trend of improvement in image quality as the training progresses. The weight corresponding to the lowest FID score was utilized to finalize the model, ensuring optimal performance on the test dataset in terms of image quality and similarity to the real distribution.

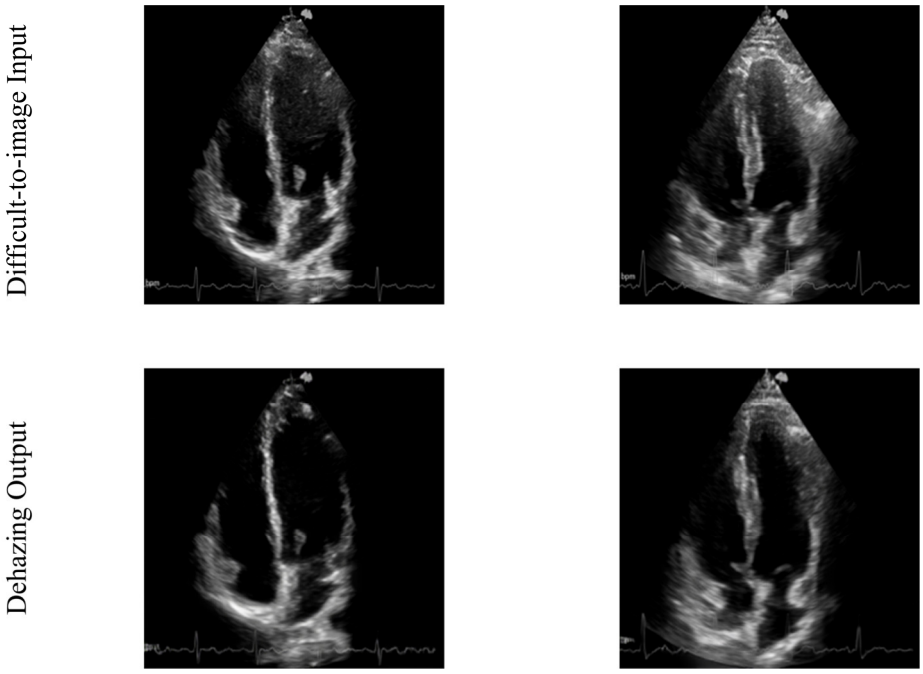

Fig. 4. The visualization of test samples on the DehazingEcho2025 dataset.

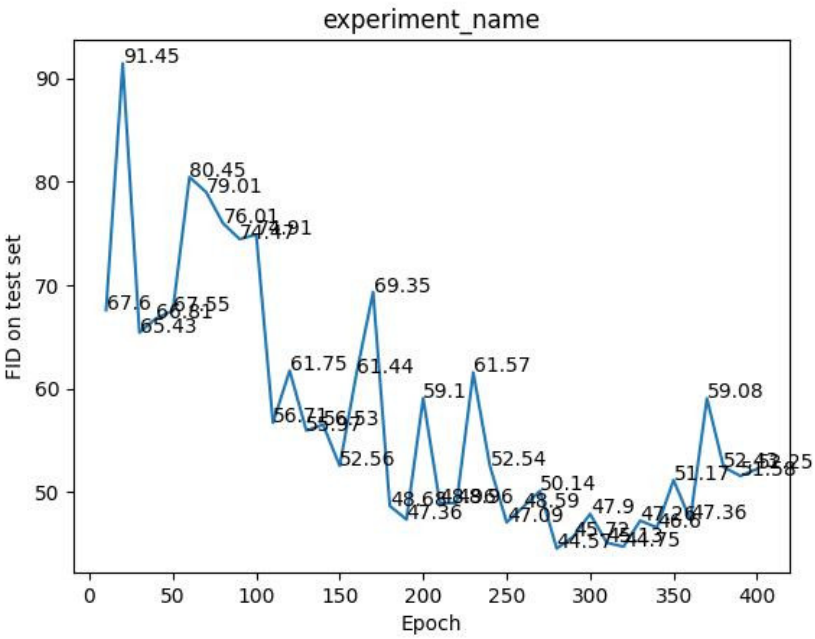

Fig. 5. The FID metric results for the test dataset.

4 Conclusion

In conclusion, the proposed 2D GAN framework, which incorporates a Mamba-enhanced Symmetric UNet as the generator and PatchGAN as the discriminator, along with a spatially-correlative loss function, demonstrates significant potential in enhancing the quality of echocardiographic images. The model's ability to effectively learn both local and global features, along with its one-sided image translation training, contributes to its robust performance in dehazing challenging echocardiography images. The experimental results on the DehazingEcho2025 dataset validate the effectiveness of our algorithm. Future work will focus on further optimizing the model and extending its application to other types of medical imaging.

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